

DCPO: DYNAMIC CLIPPING POLICY OPTIMIZATION

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ABSTRACT

Reinforcement Learning from Verifiable Rewards (RLVR) is an effective framework for improving the reasoning performance of large language models, but the empirical gains of current GRPO-style methods are limited by two measurable sources of lost signal: zero-gradient responses and excessive token clipping. These losses arise from fixed token-level probability-ratio bounds and stepwise standardization over identical rewards, which reduce the share of generated responses and tokens that can contribute to policy updates. We propose **Dynamic Clipping Policy Optimization (DCPO)**, which combines probability-aware dynamic clipping with smooth advantage standardization across cumulative training steps. The dynamic clipping rule assigns wider admissible ratios to lower-prior-probability tokens, improving token-level exploration, while the smoothing rule increases response-level utilization of generated samples. Across four mathematical reasoning benchmarks and four model scales, DCPO delivers the strongest overall empirical profile among the compared methods. On Qwen2.5-Math-7B, DCPO reaches 46.7 Avg@1 with greedy decoding and 38.8 Avg@32 with 32 samples on AIME24, compared with DAPO (36.7/31.6), GRPO (36.7/32.1), and GSPO (40.0/34.9). On Qwen2.5-14B for AIME25, DCPO obtains (23.3/19.0), exceeding GRPO (13.3/10.5), DAPO (20.0/15.3), and GSPO (16.7/9.9). Beyond final accuracy, DCPO improves the average nonzero-advantage response ratio by 28% over GRPO across four models, doubles training efficiency relative to DAPO, and lowers the token clipping ratio by an order of magnitude compared with both GRPO and DAPO. Together, these comparisons show that DCPO converts a larger fraction of the existing generated data into useful optimization signal while achieving superior benchmark performance.

1 INTRODUCTION

Recent progress in large language models (LLMs) has made reasoning performance a central empirical target, especially on tasks where correctness can be measured exactly (OpenAI, 2025; Guo et al., 2025; Comanici et al., 2025). Reinforcement Learning from Verifiable Rewards (RLVR) is attractive in this setting because it converts rule-based outcome checks into quantitative optimization signals. In domains such as mathematics and coding, where final answers can be verified, RLVR provides a direct way to compare policies by measured task accuracy and to improve models through reward-guided updates (DeepSeek-AI et al., 2024).

The GRPO family of algorithms (DeepSeek-AI et al., 2024; Liu et al., 2025; Yuan et al., 2025; Zheng et al., 2025) has become a standard RLVR baseline, and its performance has been validated across multiple studies. However, recent analyses, including DAPO (Yu et al., 2025), show that the same training pipeline can lose substantial empirical signal during optimization. In later training stages, advantages often decay to zero, creating zero-gradient responses; at the token level, fixed probability-ratio clipping can remove rarely sampled but informative tokens. Both effects reduce the measurable utilization of generated data and narrow the policy’s effective exploration space.

DAPO addresses part of this evidence gap with Clip-Higher, which relaxes the upper clipping bound, and Dynamic Sampling, which filters uniform-reward samples so that each retained batch contains valid gradients. These choices improve the comparison against GRPO but leave two quantitative limitations. (i) **Response-Level Inefficiency**: Dynamic sampling increases the number of generated responses needed for the same number of parameter updates, slowing convergence in wall-clock and GPU-hour terms. (ii) **Token Clipping Deficit**: Our analysis shows that the useful clipping range depends on the old probability of each token, so a uniform upper bound remains suboptimal for

low-probability tokens. Even methods that avoid directly discarding zero-gradient responses therefore leave measurable utilization and exploration margins unaddressed.

We introduce **Dynamic Clipping Policy Optimization (DCPO)** to improve these utilization metrics without changing the underlying RLVR supervision signal. First, DCPO uses a **smoothing technique that standardizes advantages across cumulative steps**, so responses with identical rewards at the current step can still receive nonzero learning signal from the accumulated reward distribution. This design directly targets response-level evidence loss by combining the current generated distribution with all previously generated responses for the same prompt (see Section 2). Second, DCPO replaces fixed clipping with a **dynamic clipping method that adaptively adjusts the clipping bounds based on the old token-specific probabilities**. Lower-prior-probability tokens receive wider admissible ratio ranges, increasing token-level exploration precisely where fixed clipping is most restrictive (see Section 2 and Appendix A.6). Finally, DCPO computes the mean loss only across tokens within each individual response, rather than averaging over the whole batch, preserving the relative advantage ordering among responses to the same prompt and avoiding batch-level dilution.

Empirically, we train on a merged dataset combining Math-DAPO-17k (Yu et al., 2025) with levels 3-5 of MATH (Liu et al., 2025), and evaluate on four standard mathematical reasoning benchmarks. The comparison spans four model scales and uses GRPO, DAPO, and, where available, GSPO as baselines. DCPO achieves the most favorable performance-utilization trade-off across these conditions, with the largest margins on the challenging AIME benchmarks: for AIME24-Avg@32 with 32 sampled responses, DCPO-7B scores 38.8 compared with 32.1 (GRPO), 31.6 (DAPO), and 34.9 (GSPO); for AIME25-Avg@32, DCPO-14B reaches 19.0 compared with 10.5 (GRPO), 15.3 (DAPO), and 9.9 (GSPO). The same pattern appears in training diagnostics: DCPO raises the average nonzero-advantage ratio by 28% over GRPO, doubles training efficiency relative to DAPO, and reduces token clipping by an order of magnitude compared with the baselines while also improving final benchmark scores.

Appendix A.2 provides the baseline loss definitions and advantage calculations for GRPO, DAPO, and GSPO, enabling a direct comparison between the optimization components used by prior methods and the quantitative changes introduced by DCPO.

2 DYNAMIC-ADAPTIVE CLIPPING BOUNDS FOR RL(DAC)

Probability-ratio clipping is a standard way to make reinforcement-learning updates comparable and stable across optimization steps: it limits how far the new policy probability $p(x)$ can move from the old policy probability $q(x)$. This control is especially important in off-policy importance sampling, where the reweighting ratio $r(x)$ yields an unbiased estimate of $\mathbb{E}_{q(x)}[f(x) \frac{p(x)}{q(x)}]$ but can also increase estimator variance. The variance gap in Equation (1) makes this trade-off explicit and motivates measuring clipping not only as a stability device but also as a source of discarded update signal.

$$\text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] - \text{Var}_{x \sim p} [f(x)] = \mathbb{E}_{x \sim p} \left[f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) \right] = \int f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) p(x) dx \quad (1)$$

Prior work, including PPO Schulman et al. (2017), controls this variance by constraining every token to the same fixed probability-ratio interval determined by ϵ , as in Equation (2).

$$|r(x) - 1| \leq \epsilon \quad (2)$$

The quantitative weakness of a symmetric fixed interval is that it allocates the same relative range to all tokens, even though low-probability tokens have much less absolute probability mass to move. As a result, rare tokens—the tokens most relevant to exploration diversity—are constrained most strongly in practice. DAPO partially changes the comparison by using asymmetric clipping bounds, but the bounds remain fixed rather than token-probability dependent. GSPO shifts clipping to the sequence level; when a sequence-level importance ratio falls outside the bound, an entire response can be discarded, which removes informative tokens together with high-variance ones and can reduce data utilization.

From the perspective of the variance–bias trade-off, the more targeted constraint is probability dependent: the admissible update should scale with the old probability of the token, as formalized in Equation (3).

$$|(r(x) - 1)p(x)| \leq \epsilon \quad (3)$$

Substituting $p(x) = r(x)q(x)$ into Equation (3) and enforcing non-negativity of the square-root term yields a closed-form interval for $r(x)$. The resulting **dynamic-adaptive clipping bounds** are parameterized by ϵ_{low} and ϵ_{high} but vary with $q(x)$, giving a direct quantitative mechanism for widening the update region when the old policy assigns lower probability, as shown in Equation (4).

$$0.5 + \frac{1}{2} \sqrt{\max \left(1 - \frac{4\epsilon_{low}}{q(x)}, 0 \right)} \leq r(x) \leq 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \quad (4)$$

This dynamic-adaptive rule makes the exploration benefit explicit: low-probability regions receive wider ratio ranges, while high-probability tokens retain tight stabilizing bounds. Following the dual clipping method (Ye et al., 2020), we cap the maximum bound at 10 for both positive and negative advantages so that the wider rare-token interval does not introduce excessive updates. Appendix A.5 and Appendix A.6 give the full derivation and compare the resulting bounds with the fixed-clipping baseline.

3 SMOOTH ADVANTAGE STANDARDIZATION(SAS)

In GRPO, DAPO, and GSPO, the advantage $\hat{A}_{new,j}^i$ for the j -th response is computed by standardizing only the rewards $R_{j=1,\dots,G}^i$ generated for the same prompt at the current step, as summarized by Equation (11) in Appendix A.3. This step-local comparison is simple, but it creates two measurable sources of instability. First, if all sampled responses at a step receive identical rewards, every standardized advantage becomes zero, so the generated responses add no gradient signal despite containing distinct reasoning trajectories. Second, high-entropy sampling can produce skewed label counts from step to step, causing standardized advantages to fluctuate sharply or even change sign, which makes the update signal less consistent.

Recent work (Yue et al., 2025) further suggests that RLVR mainly changes the likelihood of already-sampleable correct responses rather than replacing the full response distribution. Under this view, responses generated for the same prompt across training steps provide a larger empirical reward sample for estimating the prompt-level comparison. We therefore use a *cumulative* standardization, defined in Equation (5), to anchor the current-step signal against the reward statistics accumulated so far.

$$\hat{A}_{total,j}^i = \frac{(R_j^i - \mu_{total}^i)}{\sigma_{total}^i} \quad (5)$$

where the mean and variance are computed over all responses generated so far for the same prompt, increasing the comparison set beyond the current step.

Because both step-specific standardization ($\hat{A}_{new,j}^i$) and cumulative standardization ($\hat{A}_{total,j}^i$) can fluctuate, DCPO reports the comparison through two smoothed quantities, $\hat{S}A_{new,j}^i$ and $\hat{S}A_{total,j}^i$. These are weighted averages of the two standardizers, with weights that change as the step index i grows, as shown in Equation (6).

$$\hat{S}A_{new,j}^i = \frac{i-1}{i} \hat{A}_{new,j}^i + \frac{1}{i} \hat{A}_{total,j}^i, \quad \hat{S}A_{total,j}^i = \frac{1}{i} \hat{A}_{new,j}^i + \frac{i-1}{i} \hat{A}_{total,j}^i \quad (6)$$

$\hat{S}A_{new,j}^i$ increasingly emphasizes the current distribution as training progresses, capturing recent policy behavior. In contrast, $\hat{S}A_{total,j}^i$ gives greater weight to the cumulative distribution, treating the current-step deviation as a smaller adjustment within the broader empirical reward history. DCPO then selects the smoothed advantage with the smaller absolute value, as defined in Equation (7),

which keeps the direction of the comparison while reducing extreme standardized values that can destabilize training.

$$\hat{A}_j^i = \begin{cases} \hat{S}A_{new,j}^i, & \text{when } |\hat{S}A_{new,j}^i| < |\hat{S}A_{total,j}^i| \\ \hat{S}A_{total,j}^i, & \text{otherwise} \end{cases} \quad (7)$$

After a prompt has entered optimization, later responses for that prompt can continue to contribute even when the current-step rewards are identical. In such cases, the update uses the cumulative component $\frac{1}{i}\hat{A}_{total,j}^i$ rather than collapsing to zero, which directly raises response utilization and preserves learning signal that stepwise standardization would discard.

3.1 OTM LOSS

The **Token-Level Mean loss (TLM)** used by DAPO in Equation (13) equalizes token contributions across a batch, but this normalization can reverse the intended response-level comparison when response lengths differ. For example, suppose two responses to the same prompt have advantages 1 and 0.5, with lengths 500 and 1500, respectively. TLM assigns weights $\frac{500}{500+1500} = 0.25$ and $\frac{1500}{500+1500} = 0.75$, producing total contributions of $1 \times 0.25 = 0.25$ for response A and $0.5 \times 0.75 = 0.375$ for response B . Thus the lower-advantage response contributes more to the update solely because it is longer.

Conversely, the **Sequence-Level Mean loss (SLM)** used by GRPO in Equation (12) averages across the G responses and therefore scales down the already-standardized advantage by a factor tied to group size. This batch-level averaging weakens the empirical margins among responses, especially for large batches. DCPO treats the standardized advantages as the relevant response-level comparison and averages only across the tokens inside each response with $|o_i|$. This preserves the original relative ordering among responses while distributing each response’s importance uniformly across its tokens. We call the resulting objective **Only Token Mean loss (OTM)**, defined in Equation (8).

$$\mathcal{T}_{\text{DCPO}}(\theta) = \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \quad (8)$$

As in DAPO, DCPO removes the KL-divergence constraint from the loss. The lower and upper clipping limits are instead supplied by the dynamic bounds in Section 2, so the comparison against GRPO and DAPO isolates how probability-aware clipping and response-preserving averaging change the update signal.

4 EXPERIMENTAL SETUP

Models and Datasets We evaluate DCPO across four model scales and training regimes: Qwen2.5-Math-1.5B-Instruct, Qwen2.5-3B (common base), Qwen2.5-Math-7B (math base), and Qwen2.5-14B (common base) (Yang et al., 2024a;b). The training set contains approximately 25k mathematical problems, formed by merging DAPO-Math-17K (Yu et al., 2025) with levels 3-5 of the MATH subdataset (Hendrycks et al., 2021b; Liu et al., 2025). To keep evaluation conditions comparable across methods, every problem uses the same Qwen-Math template for training and testing (see Section A.8).

Evaluation We report results on four widely used mathematical reasoning benchmarks: MATH500 (Hendrycks et al., 2021a), AMC23 (Ouyang et al., 2022), AIME24 (Li et al., 2024), and AIME25 (MAA). For **MATH500**, which contains 500 problems, we report Avg@1. For the smaller **AMC23** (40 problems), **AIME24** (30 problems), and **AIME25** (30 problems) benchmarks, we report both Avg@1 and Avg@32 so that each method is compared under greedy decoding and under repeated stochastic sampling.

- **Avg@1:** This metric is the standard accuracy under greedy decoding and measures the model’s single best deterministic prediction.

- **Avg@32:** This metric averages accuracy over 32 sampled responses per problem with temperature 1.0 and top_p 1.0, measuring the robustness and stability of the learned policy distribution under sampling.

Baseline Settings We compare DCPO primarily against GRPO and DAPO, and include GSPO on Qwen2.5-Math-7B and Qwen2.5-14B where that baseline is evaluated. Appendix A.9 lists the clipping thresholds ϵ_* , codebase details, and shared hyperparameters, ensuring that reported margins are interpreted under matched training conditions.

5 RESULTS AND ANALYSIS

5.1 MAIN RESULTS

Table 1: Benchmark comparison across models and methods. Avg@1 reports greedy-decoding accuracy, Avg@32 reports mean accuracy over 32 sampled responses, boldface marks the best value in each model block, and each Δ row gives DCPO’s margin over the named baseline.

Model	MATH500 (Avg@1)	AMC23 (Avg@1/32)	AIME24 (Avg@1/32)	AIME25 (Avg@1/32)	Average (Avg@1/32)
Qwen2.5-Math-1.5B-Instruct					
base	73.6	57.5/49.4	10.0/10.0	3.3/6.1	36.1/21.8
GRPO	77.2	70.0/68.4	16.7/14.0	20.0/13.5	46.0/32.0
DAPO	76.0	80.0/70.6	20.0/13.5	14.4/12.5	46.5/32.4
DCPO	77.2	75.0/ 70.8	20.0/15.6	16.7/12.1	47.2/32.8
Δ_{GRPO}	+0.0	+5.0/+2.4	+3.3/+1.6	-3.3/-1.4	+1.2/+0.8
Δ_{DAPO}	+1.2	-5.0/ +0.2	+0.0/+2.1	+2.3/-0.4	+0.7/+0.4
Qwen2.5-3B[Base]					
base	46.4	27.5/7.8	3.3/0.1	3.3/0.7	20.1/2.9
GRPO	69.2	62.5/51.6	10.0/7.5	6.7/4.5	36.3/21.0
DAPO	72.4	57.5/54.0	10.0/8.3	6.7/3.9	36.6/23.1
DCPO	71.2	62.5/55.8	3.3/7.5	10.0/4.7	37.6/22.7
Δ_{GRPO}	+2.0	+0.0/+4.2	-6.7/ +0.0	+3.3/+0.2	+1.3/+1.7
Δ_{DAPO}	-1.2	+5.0/+1.8	-6.7/-0.8	+3.3/+0.8	+1.0/-0.4
Qwen2.5-Math-7B[Base]					
base	50.4	40.0/19.5	13.3/6.0	3.3/1.5	28.4/9.3
GRPO	81.6	77.5/75.9	36.7/32.1	16.7/16.7	53.1/41.6
DAPO	83.0	72.5/ 80.7	36.7/31.6	23.3/14.9	53.9/42.4
GSPO	84.0	80.0/78.8	40.0/34.9	16.7/16.2	55.2/43.3
DCPO	82.5	82.6/79.8	46.7/38.8	16.7/ 17.2	57.1/45.2
Δ_{GRPO}	+0.9	+5.1/+4.9	+10.0/+6.7	+0.0/+0.5	+4.0/+3.6
Δ_{DAPO}	-0.5	+10.1/-0.9	+10.0/+7.2	-6.6/ +2.3	+3.3/+2.8
Δ_{GSPO}	-1.5	+2.6/+1.0	+6.7/+3.9	+0.0/+1.0	+1.9/+1.9
Qwen2.5-14B[Base]					
base	60.8	47.5/16.4	3.3/1.3	3.3/1.1	28.7/6.3
GRPO	81.2	75.0/65.6	13.3/17.6	13.3/10.5	45.7/31.3
DAPO	83.4	87.5/85.1	16.7/16.4	20.0/15.3	51.9/38.9
GSPO	78.6	77.5/75.0	23.3/16.0	16.7/9.9	49.0/33.5
DCPO	84.6	85.0/79.9	20.0/ 18.2	23.3/19.0	53.2/39.0
Δ_{GRPO}	+3.4	+10.0/+14.3	+6.7/+0.6	+10.0/+8.5	+6.5/+7.7
Δ_{DAPO}	+1.2	-2.5/-5.2	+3.3/+1.8	+3.3/+3.7	+1.3/+0.1
Δ_{GSPO}	+6.0	+7.5/4.9	-3.3/ +2.2	+6.6/+10.1	+4.2/+5.5

Table 1 presents the main quantitative comparison among GRPO, DAPO, GSPO, and DCPO on MATH500, AMC23, AIME24, and AIME25. The table separates deterministic accuracy (Avg@1) from sampling robustness (Avg@32), and the Δ rows make the margin between DCPO and each baseline explicit. Across the four model scales, DCPO is consistently the best or competitive method on the average column, showing that its gains are not confined to a single architecture, dataset, or decoding condition.

Overall Superiority of DCPO On Qwen2.5-Math-1.5B-Instruct, DCPO records the highest average scores, (47.2/32.8), with positive margins over both GRPO (+1.2/+0.8) and DAPO (+0.7/+0.4). On Qwen2.5-3B, DCPO reaches (37.6/22.7), improving over GRPO by (+1.3/+1.7) and remaining competitive with DAPO: it is higher on Avg@1 by (+1.0) and lower on Avg@32 by only (-0.4). The margins widen at 7B: DCPO obtains (57.1/45.2), exceeding GRPO (+4.0/+3.6), DAPO (+3.3/+2.8), and GSPO (+1.9/+1.9). On Qwen2.5-14B, DCPO achieves (53.2/39.0), improving over GRPO by (+6.5/+7.7), over DAPO by (+1.3/+0.1), and over GSPO by (+4.2/+5.5). This pattern shows consistent practical gains in both greedy and stochastic decoding, with the largest average margins appearing on the larger base models.

Robustness on AIME under Sampling (Avg@32) AIME24 and AIME25 provide the clearest high-difficulty sampling comparison because each benchmark is small, competition-oriented, and sensitive to whether the policy can repeatedly sample correct reasoning traces. Under Avg@32, DCPO shows strong margins at the 7B and 14B scales. On AIME24-Avg@32, DCPO-7B scores 38.8, above GRPO (32.1), DAPO (31.6), and GSPO (34.9). On AIME25-Avg@32, DCPO-14B reaches 19.0, compared with 10.5 for GRPO, 15.3 for DAPO, and 9.9 for GSPO. These differences indicate that DCPO improves not only single-output accuracy but also the probability mass assigned to correct sampled solutions, which is practically important when multiple candidate outputs are generated.

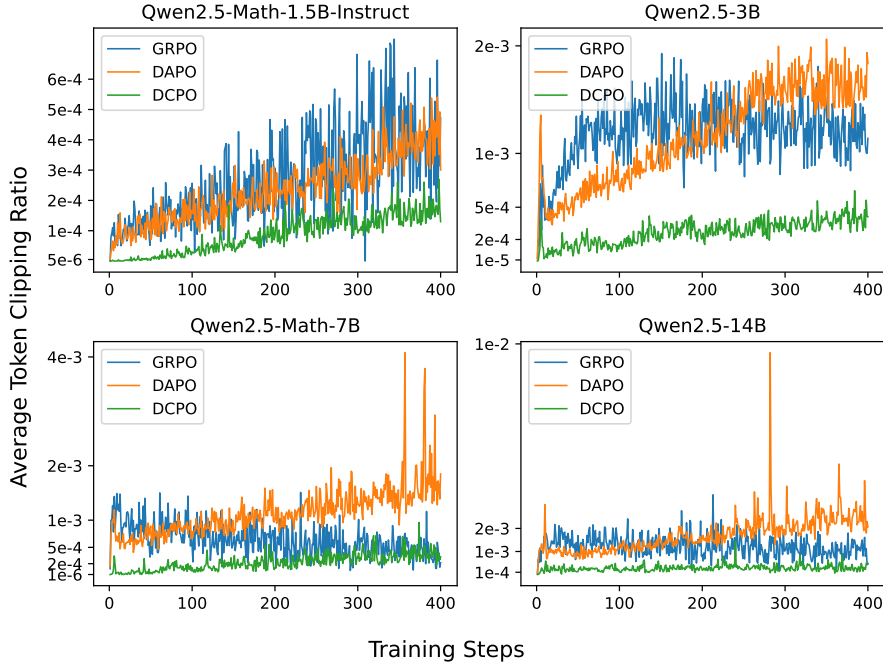


Figure 1: Token Clipping Ratio (TCR) trajectories by model and method; lower curves indicate that fewer generated tokens are removed from policy updates.

5.2 THE TOKEN CLIPPING RATIO (TCR)

We define the **Token Clipping Ratio (TCR)** as the proportion of generated tokens excluded from back-propagation by the clipping rule. This diagnostic directly measures token-level evidence loss: a lower TCR means that more tokens remain available for policy updates. Figure 1 shows that the compared methods differ substantially on this metric.

$$TCR = Average \left(\sum_{m=1}^N \frac{\text{Number of clipped tokens in } micro_m}{\text{Total number of tokens in } micro_m} \right) \quad (9)$$

where $micro_m$ denotes the m -th microbatch and N is the number of microbatches per training step.

GRPO shows model-dependent TCR trends: the ratio increases over training for Qwen2.5-Math-1.5B and Qwen2.5-3B, but decreases gradually for Qwen2.5-Math-7B and Qwen2.5-14B. DAPO, by

contrast, has an upward *TCR* trajectory across all four models, meaning that progressively more tokens are excluded as training proceeds. Both GRPO and DAPO also display large fluctuations and occasional spikes, which are unfavorable for stable optimization. GSPO has an even higher sequence-level exclusion profile on the evaluated larger models: its average *TCR* is greater than 11% on Qwen2.5-Math-7B and over 15% on Qwen2.5-14B, substantially above the average token-level clipping ratios and indicative of larger informative-token loss.

DCPO exhibits the most favorable *TCR* pattern: across model sizes and training stages, its ratio remains stable and roughly an order of magnitude below GRPO and DAPO. This low clipping rate means that DCPO retains more generated tokens for optimization while still bounding outlier probability ratios. In practical terms, the same generated batch supplies a larger usable token set, which supports the observed accuracy gains without relying on additional sampled data.

For Qwen2.5-Math-7B, the token distribution rapidly concentrates on high-confidence tokens: after roughly 60 steps, about 95% of generated tokens have old probability ($q(x) > 0.9$), and after roughly 100 steps the fraction rises to about 97% and continues increasing. As shown in Figure 4d, when the old probability satisfies ($q(x) \geq \frac{1}{1+\epsilon_{grpo}} \approx 0.83$), the new probability $p(x)$ remains inside the update interval $[\frac{1}{1+\epsilon}, 1]$ and is not discarded under the clipping rules. This explains why most tokens contribute under all methods. The decisive comparison is therefore in the low-probability tail: Figure 1 shows later-stage *TCR* growth for most training runs, indicating that low-probability tokens are increasingly at risk of being clipped and that rare-token exploration becomes the limiting factor.

Prior work (Wang et al., 2025) identifies **high-entropy tokens**—the low-probability portion of the distribution—as important contributors to emergent reasoning behavior. Because these tokens carry higher informational content, clipping them disproportionately reduces the model’s opportunity to learn diverse reasoning patterns. A wider admissible update range for low-probability tokens therefore has practical significance: it increases the effective exploration space exactly where the empirical signal is most fragile.

These diagnostics support the design of **dynamic-adaptive clipping**: DCPO relaxes clipping for tokens with $q(x) < \frac{1}{1+\epsilon}$ while preserving the safe interval for high-confidence tokens. The resulting metric pattern—low, stable *TCR* together with stronger benchmark accuracy—indicates that probability-aware clipping retains more rare-token signal without sacrificing update control.

5.3 THE RESPONSE UTILIZATION RATIO (*RUR*)

During RLVR training, a generated response affects the policy only when it has a nonzero advantage and therefore produces a nonzero gradient. We quantify response-level data efficiency with the **Response Utilization Ratio** (*RUR*), the percentage of generated responses with nonzero advantage that participate in the policy update.

$$RUR = \frac{\text{Number of responses with non-zero advantage}}{\text{Total number of generated responses}} \times 100\% \quad (10)$$

Table 2 shows that GRPO uses less than half of its generated responses on average, with an overall mean of 43.8% across the four models. GSPO is similar on the two evaluated larger models, averaging 45.6%. DCPO raises the average *RUR* to 71.8%, an absolute gain of 28% over GRPO and a relative improvement of 64%. This margin is a direct measure of DCPO’s response-level efficiency: substantially more of the already generated data contributes to parameter updates.

Table 2: Average Response Utilization Ratio (*RUR*) over 400 training steps; higher values mean more generated responses have nonzero advantages and contribute to updates.

model	GRPO	GSPO	DCPO
Qwen2.5-Math-1.5B-Instruct	45.6%	-	67.1%
Qwen2.5-3B	48.3%	-	74.3%
Qwen2.5-Math-7B	37.4%	43.5%	73.2%
Qwen2.5-14B	43.9%	47.6%	72.4%
Average	43.8%	45.6%	71.8%

Figure 2 shows that GRPO’s *RUR* falls sharply over training for every model type and size, dropping from above 90% to below 50%. On Qwen2.5-Math-7B, the ratio reaches roughly 30% after about 200 steps, indicating that current-step standardization increasingly produces zero-gradient responses.

DCPO follows a more favorable trajectory: after the first epoch, smooth advantage standardization stabilizes *RUR* around 70% and then continues to improve with additional training.

The early *RUR* values exceed 90%, whereas from the second epoch onward they remain near 70% rather than reaching 100%. This pattern is informative because it separates easy formatting effects from the more meaningful response-utilization gains produced by SAS.

First, the initial values above 90% largely reflect early model outputs that violate the required format, receive negative rewards, and therefore inflate the nonzero-advantage count. This suggests that models quickly learn the fixed answer format, after which the more important comparison is reasoning exploration. Second, the plateau near 70% reflects two recurring zero-advantage cases: (i) nearly half of the math problems are simple enough that the model consistently solves them correctly, so excluding them from further updates is reasonable because the relevant capability is already stable; (ii) the remaining cases are consistently incorrect, either because the problems are too difficult for the current model despite correct labels or because of label noise. The DAPO-Math-17K dataset also contains some items whose labels were converted to mislabeled integers, introducing potential "incorrect" labels. Keeping zero-advantage responses out of the update in these cases acts as a useful filter and reduces the risk of optimizing toward noisy or mislabeled supervision.

DAPO discards generated responses with identical labels, so its efficiency is better measured by the number of responses and training time required to reach the same number of update steps. To compensate for low sampling efficiency, we set DAPO's generation batch size to 1–4 times the training batch size and adjust it dynamically. Across the four models, DAPO requires 3 to 5 times more generated responses than DCPO for the same number of parameter updates, which at least doubles training GPU hours without improving the final performance comparison. The efficiency gap also grows with model size: on Qwen2.5-14B, DAPO uses less than 30% of responses after about 300 steps. These measurements show that DCPO's higher *RUR* translates into practical training efficiency, not just a diagnostic improvement.

Overall, the *RUR* analysis explains an important part of the accuracy table: DCPO improves benchmark performance while using a larger fraction of generated responses and requiring fewer generated samples than DAPO for equivalent update counts.

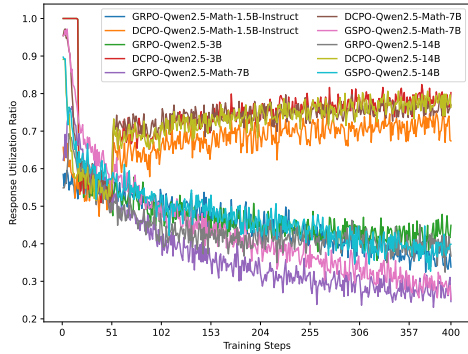


Figure 2: Response Utilization Ratio (*RUR*) during training; DCPO maintains a higher fraction of nonzero-advantage responses than GRPO.

5.4 ABLATION STUDIES

We ablate DCPO on Qwen2.5-Math-7B using *Avg@32*, the metric most directly tied to sampling robustness and policy-distribution stability. Each ablation modifies one component of the GRPO framework while keeping all other settings fixed; the KL-divergence term is removed to match DAPO, GSPO, and full DCPO. Figure 3 reports the learning curves from 20 to 400 steps, making the component-wise performance margins visible over training time.

The component curves show consistent improvement rather than isolated late-stage jumps. Each individual modification surpasses GRPO and reaches or exceeds the DAPO and GSPO range, while the combined DCPO curve gives the strongest and most stable *Avg@32* profile. The contrast is clearest in later training: baseline curves plateau or fluctuate, whereas DCPO continues upward

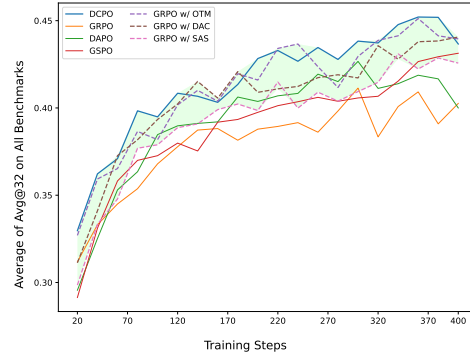


Figure 3: Ablation on Qwen2.5-Math-7B using average *Avg@32*; curves compare the contribution of each DCPO component under the same sampling metric.

both for individual components and for the full method. This pattern indicates that OTM, SAS, and DAC each contribute measurable gains, and their combination produces cumulative improvements in performance and stability.

(i) **Only-Token-Mean loss (OTM)**: Averaging the loss within each response rather than across the full batch improves the GRPO baseline, indicating that preserving response-level advantage margins is beneficial. (ii) **Smoothed Advantage Standardization (SAS)**: Replacing stepwise advantages with cumulative smoothed advantages substantially improves GRPO and approaches DAPO, showing that reward statistics accumulated across steps stabilize optimization and improve final Avg@32. (iii) **Dynamic Adaptive Clipping (DAC)**: Replacing GRPO’s fixed clipping interval with probability-adaptive bounds yields strong gains; DAC alone surpasses GRPO, DAPO, and GSPO and is more stable than OTM alone. (iv) **Combination of OTM + SAS + DAC (full DCPO)**: Combining all three components gives the best and most stable curve, demonstrating that response-level utilization, smoothed advantage estimation, and rare-token clipping each add complementary quantitative value.

Taken together, the ablation results connect the final benchmark margins to specific mechanisms: OTM preserves response-level comparisons, SAS raises usable response signal, and DAC reduces harmful token clipping. Their combined effect explains why full DCPO achieves the strongest Avg@32 trajectory on Qwen2.5-Math-7B.

6 CONCLUSION

This paper reframes RLVR policy optimization around measurable utilization of generated evidence. DCPO addresses two limitations of existing methods: fixed clipping restricts low-probability token exploration, and current-step reward standardization creates many zero-gradient responses. The proposed pipeline combines (i) **dynamically adaptive token-level clipping**, which sets probability-aware bounds from old policy probabilities and enlarges the update range for rare tokens, with (ii) **Cumulative-Smooth Advantage Standardization**, which uses reward statistics accumulated across optimization steps to preserve response-level learning signal. Across four mathematical reasoning benchmarks and four model scales, DCPO achieves the strongest overall comparison. For example, DCPO-7B reaches 38.8 Avg@32 on AIME24, compared with 32.1 for GRPO and 31.6 for DAPO, while DCPO-14B reaches 19.0 Avg@32 on AIME25. The diagnostic metrics support the same conclusion: DCPO improves response utilization by 28% over GRPO, doubles training efficiency relative to DAPO, and reduces token clipping by an order of magnitude. These results show that DCPO improves both final reasoning accuracy and the practical efficiency with which RLVR uses generated samples. Future work will evaluate whether the same utilization and clipping advantages transfer to code generation and semantic reasoning tasks.

REFERENCES

- Gheorghe Comanici, Eric Bieber, Mike Schaekermann, Ice Pasupat, Naveen Sachdeva, Inderjit Dhillon, Marcel Blistein, Ori Ram, Dan Zhang, Evan Rosen, et al. Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities. *arXiv preprint arXiv:2507.06261*, 2025.
- DeepSeek-AI, Aixin Liu, Bei Feng, Bing Xue, Bingxuan Wang, Bochao Wu, Chengda Lu, Chenggang Zhao, Chengqi Deng, Chenyu Zhang, Chong Ruan, Damai Dai, Daya Guo, Dejian Yang, Deli Chen, Dongjie Ji, Erhang Li, Fangyun Lin, Fucong Dai, Fuli Luo, Guangbo Hao, Guanting Chen, Guowei Li, H. Zhang, Han Bao, Hanwei Xu, Haocheng Wang, Haowei Zhang, Honghui Ding, Huajian Xin, Huazuo Gao, Hui Li, Hui Qu, J. L. Cai, Jian Liang, Jianzhong Guo, Jiaqi Ni, Jiashi Li, Jiawei Wang, Jin Chen, Jingchang Chen, Jingyang Yuan, Junjie Qiu, Junlong Li, Junxiao Song, Kai Dong, Kai Hu, Kaige Gao, Kang Guan, Kexin Huang, Kuai Yu, Lean Wang, Lecong Zhang, Lei Xu, Leyi Xia, Liang Zhao, Litong Wang, Liyue Zhang, Meng Li, Miaojuan Wang, Mingchuan Zhang, Minghua Zhang, Minghui Tang, Mingming Li, Ning Tian, Panpan Huang, Peiyi Wang, Peng Zhang, Qiancheng Wang, Qihao Zhu, Qinyu Chen, Qiushi Du, R. J. Chen, R. L. Jin, Ruiqi Ge, Ruisong Zhang, Ruizhe Pan, Runji Wang, Runxin Xu, Ruoyu Zhang, Ruyi Chen, S. S. Li, Shanghao Lu, Shangyan Zhou, Shanhua Chen, Shaoqing Wu, Shengfeng Ye, Shengfeng Ye, Shirong Ma, Shiyu Wang, Shuang Zhou, Shuiping Yu, Shunfeng Zhou, Shuting Pan, T. Wang, Tao Yun, Tian Pei, Tianyu Sun, W. L. Xiao, Wangding Zeng, Wanbiao Zhao, Wei An, Wen Liu,

- Wenfeng Liang, Wenjun Gao, Wenqin Yu, Wentao Zhang, X. Q. Li, Xiangyue Jin, Xianzu Wang, Xiao Bi, Xiaodong Liu, Xiaohan Wang, Xiaojin Shen, Xiaokang Chen, Xiaokang Zhang, Xiaosha Chen, Xiaotao Nie, Xiaowen Sun, Xiaoxiang Wang, Xin Cheng, Xin Liu, Xin Xie, Xingchao Liu, Xingkai Yu, Xinnan Song, Xinxia Shan, Xinyi Zhou, Xinyu Yang, Xinyuan Li, Xuecheng Su, Xuheng Lin, Y. K. Li, Y. Q. Wang, Y. X. Wei, Y. X. Zhu, Yang Zhang, Yanhong Xu, Yanhong Xu, Yanping Huang, Yao Li, Yao Zhao, Yaofeng Sun, Yaohui Li, Yaohui Wang, Yi Yu, Yi Zheng, Yichao Zhang, Yifan Shi, Yiliang Xiong, Ying He, Ying Tang, Yishi Piao, Yisong Wang, Yixuan Tan, Yiyang Ma, Yiyuan Liu, Yongqiang Guo, Yu Wu, Yuan Ou, Yuchen Zhu, Yudian Wang, Yue Gong, Yuheng Zou, Yujia He, Yukun Zha, Yunfan Xiong, Yunxian Ma, Yuting Yan, Yuxiang Luo, Yuxiang You, Yuxuan Liu, Yuyang Zhou, Z. F. Wu, Z. Z. Ren, Zehui Ren, Zhangli Sha, Zhe Fu, Zhean Xu, Zhen Huang, Zhen Zhang, Zhenda Xie, Zhengyan Zhang, Zhewen Hao, Zhibin Gou, Zhicheng Ma, Zhigang Yan, Zhihong Shao, Zhipeng Xu, Zhiyu Wu, Zhongyu Zhang, Zhuoshu Li, Zihui Gu, Zijia Zhu, Zijun Liu, Zilin Li, Ziwei Xie, Ziyang Song, Ziyi Gao, and Zizheng Pan. Deepseek-v3 technical report, 2024. URL <https://arxiv.org/abs/2412.19437>.
- Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, et al. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.
- Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the math dataset. *arXiv preprint arXiv:2103.03874*, 2021a.
- Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the math dataset. *Sort*, 2(4): 0–6, 2021b.
- Jia Li, Edward Beeching, Lewis Tunstall, Ben Lipkin, Roman Soletskyi, Shengyi Huang, Kashif Rasul, Longhui Yu, Albert Q Jiang, Ziju Shen, et al. Numinamath: The largest public dataset in ai4maths with 860k pairs of competition math problems and solutions. *Hugging Face repository*, 13:9, 2024.
- Zichen Liu, Changyu Chen, Wenjun Li, Penghui Qi, Tianyu Pang, Chao Du, Wee Sun Lee, and Min Lin. Understanding r1-zero-like training: A critical perspective. *arXiv preprint arXiv:2503.20783*, 2025.
- MAA. American invitational mathematics examination (AIME). Mathematics Competition Series, 2024. URL <https://maa.org/math-competitions/aime>.
- OpenAI. Introducing openai o3 and o4-mini. 2025. URL <https://openai.com/index/introducing-o3-and-o4-mini/>.
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. *Advances in neural information processing systems*, 35:27730–27744, 2022.
- John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.
- Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, YK Li, Yang Wu, et al. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.
- Guangming Sheng, Chi Zhang, Zilingfeng Ye, Xibin Wu, Wang Zhang, Ru Zhang, Yanghua Peng, Haibin Lin, and Chuan Wu. Hybridflow: A flexible and efficient rlhf framework. *arXiv preprint arXiv:2409.19256*, 2024.
- Shenzhi Wang, Le Yu, Chang Gao, Chujie Zheng, Shixuan Liu, Rui Lu, Kai Dang, Xionghui Chen, Jianxin Yang, Zhenru Zhang, et al. Beyond the 80/20 rule: High-entropy minority tokens drive effective reinforcement learning for llm reasoning. *arXiv preprint arXiv:2506.01939*, 2025.

- An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, et al. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024a.
- An Yang, Beichen Zhang, Binyuan Hui, Bofei Gao, Bowen Yu, Chengpeng Li, Dayiheng Liu, Jianhong Tu, Jingren Zhou, Junyang Lin, et al. Qwen2.5-math technical report: Toward mathematical expert model via self-improvement. *arXiv preprint arXiv:2409.12122*, 2024b.
- Deheng Ye, Zhao Liu, Mingfei Sun, Bei Shi, Peilin Zhao, Hao Wu, Hongsheng Yu, Shaojie Yang, Xipeng Wu, Qingwei Guo, et al. Mastering complex control in moba games with deep reinforcement learning. In *Proceedings of the AAAI conference on artificial intelligence*, volume 34, pp. 6672–6679, 2020.
- Qiyang Yu, Zheng Zhang, Ruofei Zhu, Yufeng Yuan, Xiaochen Zuo, Yu Yue, Tiantian Fan, Gaohong Liu, Lingjun Liu, Xin Liu, et al. Dapo: An open-source llm reinforcement learning system at scale. *arXiv preprint arXiv:2503.14476*, 2025.
- Yufeng Yuan, Qiyang Yu, Xiaochen Zuo, Ruofei Zhu, Wenyuan Xu, Jiaze Chen, Chengyi Wang, Tiantian Fan, Zhengyin Du, Xiangpeng Wei, et al. Vapo: Efficient and reliable reinforcement learning for advanced reasoning tasks. *arXiv preprint arXiv:2504.05118*, 2025.
- Yang Yue, Zhiqi Chen, Rui Lu, Andrew Zhao, Zhaokai Wang, Shiji Song, and Gao Huang. Does reinforcement learning really incentivize reasoning capacity in llms beyond the base model? *arXiv preprint arXiv:2504.13837*, 2025.
- Chujie Zheng, Shixuan Liu, Mingze Li, Xiong-Hui Chen, Bowen Yu, Chang Gao, Kai Dang, Yuqiong Liu, Rui Men, An Yang, et al. Group sequence policy optimization. *arXiv preprint arXiv:2507.18071*, 2025.

A APPENDIX

A.1 MAIN SYMBOL DEFINITIONS

G	Number of responses generated per prompt at each step.
θ	Parameters of the actor model.
$\pi_{\theta}(o_i q)$	Probability of response o_i given prompt q under parameters θ .
$\pi_{\theta_{old}}(o_i q)$	Probability of response o_i given prompt q under previous parameters θ_{old} .
π_{ref}	Reference policy for KL divergence computation.
β	Coefficient for KL divergence term (controls policy update vs. constraint trade-off).
$r_{i,t}(\theta)$	Probability ratio $\frac{\pi_{\theta}(o_i^t q, o_i^{1:t-1})}{\pi_{\theta_{old}}(o_i^t q, o_i^{1:t-1})}$ for token t in response o_i .
$\hat{A}_{i,t}$	Standardized advantage estimate for token t in response o_i .
$ o_i $	The length of the response in tokens.
$\mathbb{D}_{KL}(\pi_{\theta} \pi_{ref})$	KL divergence between current and reference policies.
ε	Probability ratio clipping threshold.
ε_{low} and ε_{high}	Lower/upper thresholds for probability ratio clipping.
M	Maximum allowed response length.
$is_equivalent(a, o_i)$	Binary function indicating whether response o_i matches answer a .
μ_{new}^i	Mean reward across G responses generated at step i (0 if $i = 0$).
σ_{new}^i	Reward variance across G responses at step i (0 if $i = 0$).
μ_{old}^i	Mean reward across $G \cdot (i - 1)$ responses from steps 1 to $i - 1$ (0 if $i \leq 1$).
σ_{old}^i	Reward variance across $G \cdot (i - 1)$ responses from steps 1 to $i - 1$ (0 if $i \leq 1$).
μ_{total}^i	Mean reward across $G \cdot i$ responses from steps 1 to i (0 if $i = 0$).
σ_{total}^i	Reward variance across $G \cdot i$ responses from steps 1 to i (0 if $i = 0$).

A.2 PRELIMINARY

A.3 ADVANTAGE CALCULATION

In GRPO and DAPO, the token advantage $\hat{A}_{j,t}^i$ for token t in response j is computed by comparing reward R_j^i with the mean μ^i and standard deviation σ^i of the G responses generated for the same prompt at step i , as shown in Equation (11). This makes each advantage a within-step quantitative comparison.

$$\hat{A}_{j,t}^i = \frac{(R_j^i - \mu^i)}{\sigma^i} \quad (11)$$

When all responses for a prompt receive identical rewards, this within-step comparison assigns zero advantage to every response, so none of those generated samples contributes to the update.

A.3.1 GRPO

Shao et al. (2024) proposed the **Group Relative Policy Optimization (GRPO)**, which samples G responses for each query, scores them with a rule-based reward function, and estimates token-level advantages using the within-step comparison in Equation (11). The policy is then updated with Equation (12), referred to as **Sequence Level Mean loss (SLM)**:

$$\mathcal{T}_{\text{GRPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_{i,t} \right) - \beta \mathbb{D}_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}}) \quad (12)$$

The first term averages token-level advantages within each response and then across all G responses in the batch. The $\mathbb{D}_{\text{KL}}$ penalty limits deviation from a reference policy (e.g., a pre-trained model), and the fixed threshold ε bounds token probability-ratio changes. This widely used formulation provides a strong baseline, but its batch-level averaging can dilute the empirical importance of individual responses.

A.3.2 DAPO

Yu et al. (2025) proposed **Dynamic sAmpling Policy Optimization (DAPO)** after observing that GRPO can give longer sequences disproportionate influence. DAPO normalizes by total token count across the batch, producing the **Token Level Mean loss (TLM)** in Equation (13).

$$\mathcal{T}_{\text{DAPO}}(\theta) = \frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \quad (13)$$

subject to $0 < |\{o_i \mid \text{is_equivalent}(a, o_i)\}| < G$. Prompts whose responses all share the same reward are discarded and regenerated to maintain batch size, which improves the retained gradient signal but increases the number of generated responses required for the same number of updates.

A.3.3 GSPO

Zheng et al. (2025) proposed **Group Sequence Policy Optimization (GSPO)**, arguing that token-level clipping can add high-variance noise to gradient estimates, especially over long sequences. GSPO therefore compares and clips at the sequence level, replacing token-wise clipping with a sequence-wise ratio intended to produce a more stable training signal.

$$\mathcal{T}_{\text{GSPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \min \left(s_i(\theta) \hat{A}_i, \text{clip}(s_i(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_i \right) \quad (14)$$

where $s_i(\theta) = \exp\left(\frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \log\left(\frac{\pi_{\theta}(o_{j,t}|q)}{\pi_{\theta_{old}}(o_{j,t}|q)}\right)\right)$

In our comparison, GSPO discards more than 10% of responses with nonzero advantages, which reduces usable response evidence and training efficiency. GRPO retains more sequence-level responses, but fixed token-level clipping still leaves many high-entropy tokens exposed to unstable updates.

A.4 REWARD CALCULATION

To keep the supervision signal verifiable and comparable across methods, we use a rule-based reward model that scores each generated response by answer correctness and format compliance, as in Equation (15). A response receives +1 when both format and answer are correct, 0 when the format is correct but the answer is incorrect, and -1 when the format is incorrect.

$$R_j^i = \begin{cases} 1, & \text{correct format and correct answer} \\ 0, & \text{correct format but incorrect answer} \\ -1, & \text{incorrect format} \end{cases} \quad (15)$$

where j indexes the response group for a given problem at training step i , and the reward is determined only by the response's format and correctness.

A.5 HYPOTHETICAL REASONING PROCESS FOR DYNAMIC-ADAPTIVE CLIPPING BOUNDS

Probability-ratio clipping stabilizes reinforcement-learning updates by limiting large parameter shifts, and it is especially important when samples from an old policy are reused to estimate expectations under a new policy. The following derivation shows why a fixed ratio interval is not the only way to control this comparison.

The expected value of a function $f(x)$ under the new probability $p(x)$ can be rewritten as an expectation under the old probability $q(x)$ by applying the importance-sampling weight, as shown in equation 16:

$$\int f(x)p(x) dx = \int f(x) \frac{p(x)}{q(x)} q(x) dx = \mathbb{E}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] \quad (16)$$

Although this estimator is unbiased, its variance can be much larger than the true variance under the new policy. Equation (17) gives the true variance of $f(x)$ under $p(x)$, while Equation (18) gives the variance of the importance-sampled estimator under $q(x)$:

$$\text{Var}_{x \sim p}[f(x)] = \mathbb{E}_{x \sim p}[f(x)^2] - (\mathbb{E}_{x \sim p}[f(x)])^2 \quad (17)$$

$$\begin{aligned} \text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] &= \mathbb{E}_{x \sim q} \left[\left(f(x) \frac{p(x)}{q(x)} \right)^2 \right] - \left(\mathbb{E}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] \right)^2 \\ &= \mathbb{E}_{x \sim p} \left[f(x)^2 \frac{p(x)}{q(x)} \right] - (\mathbb{E}_{x \sim p}[f(x)])^2 \end{aligned} \quad (18)$$

The difference in Equation (19) isolates the variance inflation term that motivates clipping or other constraints on the probability ratio:

$$\text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] - \text{Var}_{x \sim p}[f(x)] = \mathbb{E}_{x \sim p} \left[f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) \right] = \int f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) p(x) dx \quad (19)$$

Many prior methods, including PPO Schulman et al. (2017) and RLVR algorithms (DeepSeek-AI et al., 2024; Liu et al., 2025), reduce this inflation by applying a fixed symmetric bound to the probability ratio $r(x) = \frac{p(x)}{q(x)}$, as defined in Equation (20):

$$|r(x) - 1| \leq \epsilon \quad (20)$$

However, the fixed interval gives the same relative allowance to every token, even though the practical exploration margin differs with $q(x)$. When the old policy assigns low probability mass, the absolute update region is small, so rare tokens are restricted more strongly. DAPO’s asymmetric bounds change the upper and lower constants, but they still do not adapt to the token-specific old probability.

Recent studies such as Yue et al. (2025) suggest that the reasoning paths used by RLVR models are largely already present in the base model’s sampling distribution. The small fraction of clipped parameters during off-policy batch updates is consistent with this relatively modest policy shift. We therefore replace the conventional fixed constraint in Equation (20) with the dynamic-adaptive restriction in Equation (21), which includes the probability term directly in the bound.

$$\left| \left(\frac{p(x)}{q(x)} - 1 \right) p(x) \right| \leq \epsilon \quad (21)$$

Substituting $p(x) = r(x)q(x)$ into the preceding inequality yields the expression in Equation (22).

$$-\epsilon \leq (r(x) - 1)r(x) * q(x) \leq \epsilon \quad (22)$$

For this clipping ratio to be mathematically valid and meaningful, we must adhere to the fundamental constraints of probability distributions and their ratios. Specifically, as shown in Equation (23), the probabilities $p(x)$ and $q(x)$ must be non-negative and less than or equal to 1. Consequently, the probability ratio $r(x)$ must also be non-negative, and the clipping hyper-parameter ϵ must be non-negative.

$$\begin{cases} 0 \leq p(x) \leq 1 \\ 0 \leq q(x) \leq 1 \\ 0 \leq r(x) \\ 0 \leq \epsilon \end{cases} \quad (23)$$

To ensure the validity and practical applicability of the clipping bound derived from Equation (22), we define the necessary conditions for its bounds as outlined in Equation (24).

$$\begin{cases} low_{ge}(q(x), \epsilon) = 0.5 - \frac{1}{2} \sqrt{1 + \frac{4\epsilon}{q(x)}} \\ high_{le}(q(x), \epsilon) = 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon}{q(x)}} \\ low_{le}(q(x), \epsilon) = 0.5 - \frac{1}{2} \sqrt{1 - \frac{4\epsilon}{q(x)}}, & \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \\ high_{ge}(q(x), \epsilon) = 0.5 + \frac{1}{2} \sqrt{1 - \frac{4\epsilon}{q(x)}}, & \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \end{cases} \quad (24)$$

As we can get from Equation (22) to Equation (24), the probability ratio $r(x)$ should be met by the following conditions:

$$\begin{cases} low_{ge}(q(x), \epsilon) \leq 0 \leq r(x) \leq high_{le}(q(x), \epsilon), & \text{when } \text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] \geq \text{Var}_{x \sim p} [f(x)] \\ r(x) \leq low_{le}(q(x), \epsilon) \cup high_{ge}(q(x), \epsilon) \leq r(x), & \text{otherwise} \end{cases} \quad (25)$$

For the clipping bound, we consider the overlapping region centered around the ratio value of 1 to minimize the discrepancy of bounds between the two conditions and to maintain consistency with the conventional fixed bound of GRPO, and define the following conditions to ensure the validity of the clipping bound:

$$\begin{aligned} high_{ge}(q(x), \epsilon) \leq r(x) \leq high_{le}(q(x), \epsilon) \\ \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \end{aligned} \quad (26)$$

For low_{le} and $high_{ge}$, the square-root term must remain nonnegative, so we use the maximum with zero, e.g., $\max(1 - \frac{4\epsilon}{q(x)}, 0)$. This yields the following **dynamic-adaptive clipping bounds for importance sampling**:

$$0.5 + \frac{1}{2} \sqrt{\max\left(1 - \frac{4\epsilon_{low}}{q(x)}, 0\right)} \leq r(x) \leq 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \quad (27)$$

For the upper and lower bounds in Inequality 27, we choose separate hyperparameters ϵ_{high} and ϵ_{low} so that the adaptive interval matches GRPO at selected reference points while varying with $q(x)$ elsewhere.

A.6 ANALYZING THE DIFFERENCES BETWEEN DYNAMIC-ADAPTIVE CLIPPING AND FIXED CLIPPING

To compare dynamic-adaptive clipping fairly with the fixed symmetric GRPO interval, we align the adaptive bounds at the boundary points $(\frac{1}{1+\epsilon}, 1)$ and $(1, 1 - \epsilon)$, as shown in Figure 4d. Solving Equation (28) gives the corresponding values of ϵ_{low} and ϵ_{high} .

$$\begin{cases} q(x) = \frac{1}{1+\epsilon} \\ 1 = 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \end{cases} \quad (28)$$

$$\begin{cases} q(x) = 1 \\ 1 - \epsilon = 0.5 + \frac{1}{2} \sqrt{\max(1 - \frac{4\epsilon_{low}}{q(x)}, 0)} \end{cases}$$

With the common GRPO setting $\epsilon = 0.2$, this alignment gives approximately $\epsilon_{low} = 0.16$ and $\epsilon_{high} = 0.2$. Although these values are asymmetric, their role differs from DAPO’s fixed asymmetric clipping: here the asymmetry calibrates a probability-dependent interval that still changes with $q(x)$.

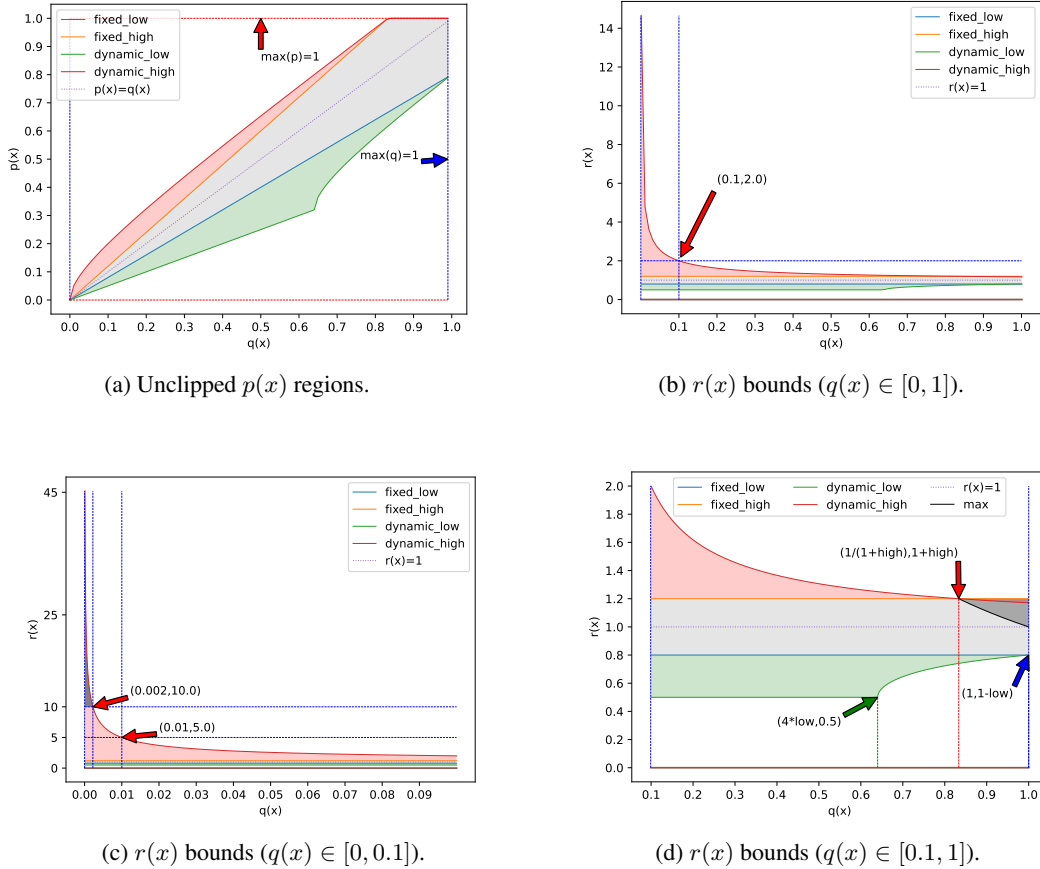


Figure 4: Fixed versus dynamic-adaptive clipping bounds. The curves compare GRPO-style fixed clipping ($\epsilon = 0.2$) with probability-dependent DCPO clipping ($\epsilon_{low} = 0.16$, $\epsilon_{high} = 0.2$), highlighting the wider admissible region for low- $q(x)$ tokens.

Dynamic versus fixed clipping: Figure 4 compares the fixed GRPO interval ($[1 - \epsilon, 1 + \epsilon]$) with probability-aware adaptive clipping. For a token with old probability $q(x)$, fixed clipping provides the same relative width 2ϵ for every token, so the absolute exploration space shrinks linearly as $q(x)$ decreases (Figure 4a). Dynamic adaptation instead widens the relative interval as $q(x)$ becomes smaller, preserving more usable exploration space in the low-probability region.

Effect of decreasing $q(x)$: When $q(x)$ becomes small, the fixed interval $[1 - \epsilon, 1 + \epsilon]$ remains unchanged, so it does not compensate for the smaller absolute probability mass of rare tokens. The dynamic-adaptive bound expands rapidly in this regime (Figure 4b), giving rare tokens a larger relative update range and improving the quantitative exploration margin.

Behavior for extremely low probabilities: Figure 4c focuses on $q(x) \in [0, 0.1]$, where the difference between the two clipping rules is largest. As $q(x) \rightarrow 0$, the dynamic-adaptive upper bound for $r(x)$

increases rapidly with the inverse square-root form, whereas the fixed-clipping value remains constant. Inspired by dual clipping (Ye et al., 2020), we impose a hard ceiling $r_{\max} = 10$ to prevent instability such as gradient vanishing or explosion. The gray region in Figure 4c corresponds to approximately $q(x) \leq 2 \times 10^{-3}$, where the adaptive upper limit is capped at $r_{\max}$.

Upper and lower bounds: For the lower bound, when $q(x) \leq 4\epsilon$, dynamic-adaptive clipping gives $dynamic_{low} = 0.5$, whereas fixed clipping keeps $fixed_{low} = 1 - \epsilon = 0.8$. As $q(x)$ increases beyond this threshold, the adaptive lower bound smoothly contracts toward the fixed value. Similarly, the adaptive upper bound is larger than the fixed upper bound whenever $q(x) < \frac{1}{1+\epsilon_{high}}$ and merges with it for larger $q(x)$. Thus, $[dynamic_{low}(q(x), \epsilon_{low}), dynamic_{high}(q(x), \epsilon_{high})]$ widens the admissible region for rare tokens while retaining the tighter fixed-like interval for common tokens (Figure 4d).

Positive versus negative advantage: The loss in Equation (8) contains both positive- and negative-advantage terms. For a positive advantage, the minimizing operation clips $r(x)$ within $[0, dynamic_{high}(q(x), \epsilon_{high})]$; for a negative advantage, it uses $[dynamic_{low}(q(x), \epsilon_{low}), \infty)$. To avoid destabilizing updates from excessively large $r(x)$, we cap the ratio at $r_{\max} = 10$ for both advantage signs, following the motivation of Ye et al. (2020).

In short, for both positive and negative advantages, dynamic-adaptive clipping preserves the stabilizing role of clipping while granting rare low-probability tokens a larger measured exploration interval than fixed clipping.

A.7 THE DEFINITIONS OF THE ADVANTAGE STANDARDIZATION

Let i denote the i -th iteration in which prompts generate responses and update model parameters. The following statistics define the reward-normalization comparisons used by SAS.

For response rewards, the current-step, historical, and cumulative means are defined in Equation (29):

$$\begin{cases} \mu_{new}^i = \frac{1}{G} \sum_{j=1}^G R_j^i \\ \mu_{old}^i = \frac{1}{G*(i-1)} \sum_{k=1}^{i-1} \sum_{j=1}^G R_j^k \\ \mu_{total}^i = \frac{1}{G*i} \sum_{k=1}^i \sum_{j=1}^G R_j^k \\ \quad = \frac{1}{i} (\mu_{new}^i + (i-1) \mu_{old}^i) \end{cases} \quad (29)$$

The corresponding current-step, historical, and cumulative standard deviations are defined in Equation (30):

$$\begin{cases} \sigma_{new}^i = \sqrt{\frac{1}{G} \sum_{j=1}^G (R_j^i - \mu_{new}^i)^2} \\ \sigma_{old}^i = \sqrt{\frac{1}{G*(i-1)} \sum_{k=1}^{i-1} \sum_{j=1}^G (R_j^k - \mu_{old}^k)^2} \\ \sigma_{total}^i = \sqrt{\frac{1}{G*i} \sum_{k=1}^i \sum_{j=1}^G (R_j^k - \mu_{total}^k)^2} \\ \quad = \sqrt{\frac{1}{i} \left(\sigma_{new}^i{}^2 + (i-1) \sigma_{old}^i{}^2 + \frac{i-1}{i} (\mu_{old}^i - \mu_{new}^i)^2 \right)} \end{cases} \quad (30)$$

To reduce resource usage when computing σ_{total}^i and μ_{total}^i in Equation (30) and Equation (29), we use equivalent calculations from μ_{new}^i , σ_{new}^i , μ_{old}^i , and σ_{old}^i rather than directly recomputing over all original rewards.

A.8 THE TEMPLATE FOR EXPERIMENTS

All training and evaluation runs use the same Qwen-Math-based template (Yang et al., 2024b), ensuring that method comparisons are not confounded by prompt-format differences.

Qwen-Math Template

```
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{ }. <|im_end|>
<|im_start|>user
{problem}
<|im_end|>
<|im_start|>assistant
```

A.9 TRAINING SETTING

For GRPO, we set the symmetric clipping threshold to $\epsilon = 0.2$. For DAPO, we use lower/upper clipping thresholds (0.2, 0.28), a 512-token soft-punish cache, and maximum probability ratio 10 for negative advantages. For GSPO, we use lower/upper clipping thresholds ($3e - 4, 4e - 4$). For DCPO, we use low/high clipping thresholds (0.16, 0.2), preserving the GRPO reference points $(q(x), r(x)) = (\frac{1}{1+\epsilon}, 1 + \epsilon)$ and $(q(x), r(x)) = (1, 1 - \epsilon)$ (Figure 4d), and set the maximum probability ratio to 10 for both positive and negative advantages.

All methods are trained for 400 steps with a response-generation batch size of 512 and a parameter-update mini-batch size of 32. We generate $G=16$ responses per problem using temperature 1 and top_p 1. The prompt maximum length is 1,024 and the response maximum length is 3,072, substantially shorter than the 20k maximum length used in the original DAPO.

We adapt the training codebase from Verl (Sheng et al., 2024) and run experiments on 32xH20 GPUs. Unless otherwise stated, hyperparameters match DAPO. For loss computation, we retain the original $\frac{1}{G}$ or $\sum_{i=1}^G \frac{1}{|o_i|}$ weighted-average formulation from GRPO or DAPO rather than Verl’s approximate weighted calculation, ensuring that comparisons remain compatible across different microbatch sizes.

A.10 THE DETAILED PERFORMANCE OF MODELS

Avg@1 Trend Figure 5 reports average Avg@1, computed by averaging greedy-decoding accuracy over MATH500, AMC23, AIME24, and AIME25. This figure isolates deterministic prediction quality during training. DCPO remains above GRPO and DAPO across all four model scales and above GSPO on the two evaluated GSPO models. The trajectory is also smoother: it avoids the pronounced GRPO performance drop and the larger DAPO fluctuations, indicating that the accuracy gains in Table 1 are supported by a more stable greedy-decoding training path.

Avg@32 Trend Figure 6 compares DCPO with GRPO, DAPO, and, where available, GSPO using Avg@32 on AMC23, AIME24, and AIME25. This metric emphasizes whether a trained policy assigns enough probability mass to correct solutions under repeated sampling. GRPO shows substantial fluctuation and occasional collapse; for Qwen2.5-14B, its Avg@32 drops noticeably after approximately 170 steps. Across all evaluated scales from 1.5B to 14B, DCPO instead follows a consistently upward trajectory with comparatively small fluctuations, including faster early-stage growth. Relative to DAPO, DCPO achieves stronger performance while using about half the training time and resources, linking the sampling-robustness gains to the efficiency results in Section 5.3.

Entropy Trend Figure 7 shows that GRPO undergoes sharp entropy decay after the initial warm-up and then remains in a low-entropy band. This collapse quantitatively reflects reduced policy randomness, fewer diverse sampled responses, and a weaker basis for subsequent exploration.

The DAPO and DCPO entropy patterns depend on model scale. On the instruct model (Qwen2.5-Math-1.5B), DAPO, DCPO, and GRPO all decline similarly. On the larger base models (Qwen2.5-3B, Qwen2.5-Math-7B, and Qwen2.5-14B), DAPO maintains the highest entropy and shows large

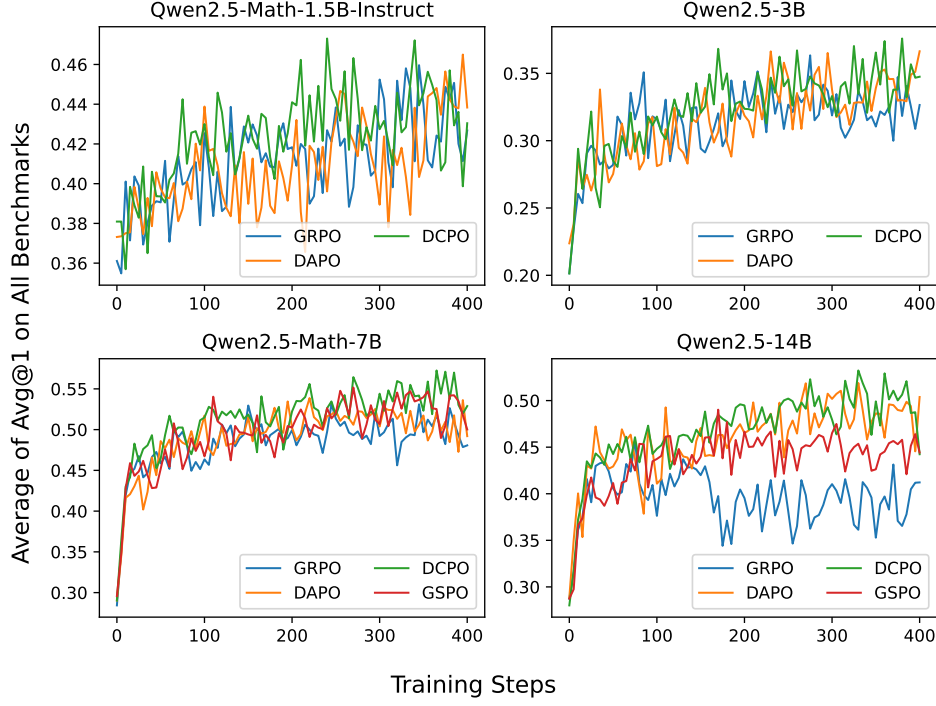


Figure 5: Average Avg@1 across MATH500, AMC23, AIME24, and AIME25; curves compare greedy-decoding accuracy over training.

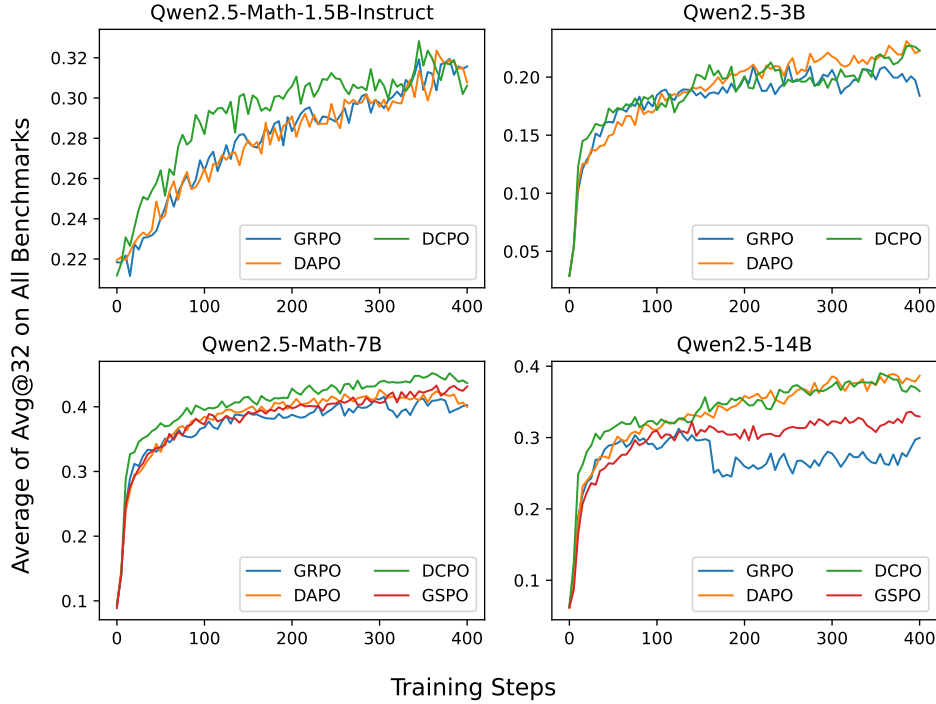


Figure 6: Average Avg@32 across AMC23, AIME24, and AIME25; curves compare robustness under 32 sampled responses per problem.

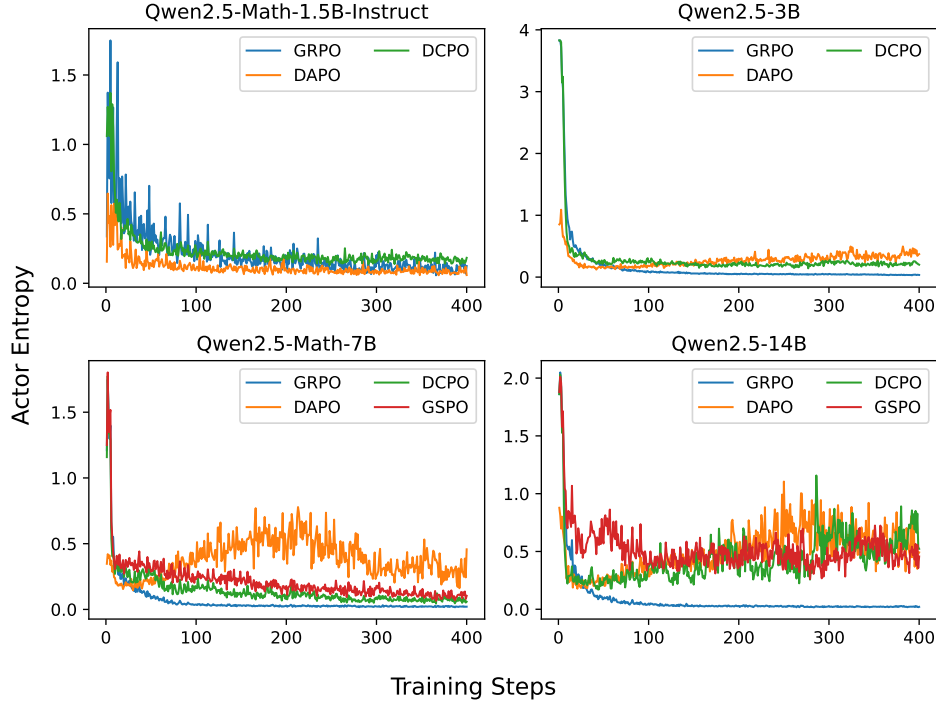


Figure 7: Training entropy by model and method; DCPO stays between GRPO’s low-entropy collapse and DAPO’s higher-variance entropy regime.

fluctuations, whereas DCPO occupies a moderate and smoother band. GSPO has entropy similar to DCPO on Qwen2.5-Math-7B, consistent with comparable average metrics there, but on Qwen2.5-14B its performance trends downward with fluctuations. DCPO therefore avoids both the low-entropy region associated with GRPO and the high-variance entropy region associated with DAPO.

These observations support a U-shaped interpretation of entropy versus performance. Too little entropy reduces exploratory signal and encourages premature convergence, while too much or highly unstable entropy weakens optimization stability. DCPO’s moderate entropy band aligns with its stronger Avg@1, Avg@32, *TCR*, and *RUR* results, suggesting that the method improves reasoning performance by balancing exploration with stable policy updates across model sizes.